

UNIVERSITAT POLITÈCNICA DE CATALUNYA

School of Professional & Executive Development

Postgrado en

Mobile Apps Design

FUNDAMENTALS

Juan Luis Gorricho

SETTING THE PROGRAMING TOOLS

- Download Android Studio,
- open up Android Studio to continue with the installation, follow the instructions as requested,
- mark the check-boxes to install:
  - HAXM (the hardware accelerator)
  - Android Virtual Device
- once installed, create a first Android Studio project accepting everything by default, if a new version of android-sdk is suggested to be installed, proceed that way,
- finally, try to run the project as it is on a Virtual Device, if you don't have any mobile emulator created yet, do it now.

ANDROID

Application components:

- **Activity**: An *activity* provides the user interface usually in the form of a single screen and handles all events triggered from that interface.
- **Service**: A *service* is a component that runs in background to perform long-term operations, periodic operations or activities involving remote connections.
- **Broadcast receiver**: is inactive until the reception of a broadcast message, reacting to the event and executing a given task.
- **Content provider**: A *content provider* manages some application data to make it available to other applications.

INTENTS

- An **Intent** object is the communication unit between components of the same or different applications. This is a passive object keeping a data structure with instructions about an operation to be executed.
- In any case, the Android OS identifies the activity, service or set of broadcast receivers to receive the intent, instantiating them if necessary.

INTENTS

- The Intent object is composed of:
  - **Component name**: naming the receiver component of the intent.
  - **Action**: string of characters indicating the action to be executed.
  - **Data**: the **URI** address location of the data to work on and the **type** (MIME type) of data to be processed.
  - **Category**: complementary to the action, string of characters indicating the type of component which will process the intent.
  - **Extras**: pairs key-value with complementary data.
  - **Flags**: auxiliary flags to warn Android how the receiving component must be executed.

INTENT FILTERS

- Declaring intent filters at the manifest file:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest . . . >
    <application . . . >
        <activity android:name="com.example.project.FreneticActivity"
                  android:icon="@drawable/small_pic.png"
                  android:label="@string/freneticLabel"
                  . . .  >
            <intent-filter . . . >
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
            <intent-filter . . . >
                <action android:name="com.example.project.BOUNCE" />
                <data android:mimeType="image/jpeg" />
                <category android:name="android.intent.category.DEFAULT" />
            </intent-filter>
        </activity>
        . . .
    </application>
</manifest>
```

STARTING AN ACTIVITY

- Creating an Intent with **explicit** resolution:

```
Intent myIntent = new Intent( this , Activity2.class );
startActivity(myIntent);
```

- Creating an Intent with **implicit** resolution:

```
Intent myIntent = new Intent();
myIntent.setAction("edu.upc.exercise2");
startActivity(myIntent);
```

ACTIVITIES AND TASKS

- Android defines the **Task** concept or “stack of activities” for a given application.
- An activity initiates another one, for example, calling startActivity(intent), being the target activity from the same or another application. Once the new activity is initiated, on pressing the “**back**” button we come back to the original activity.

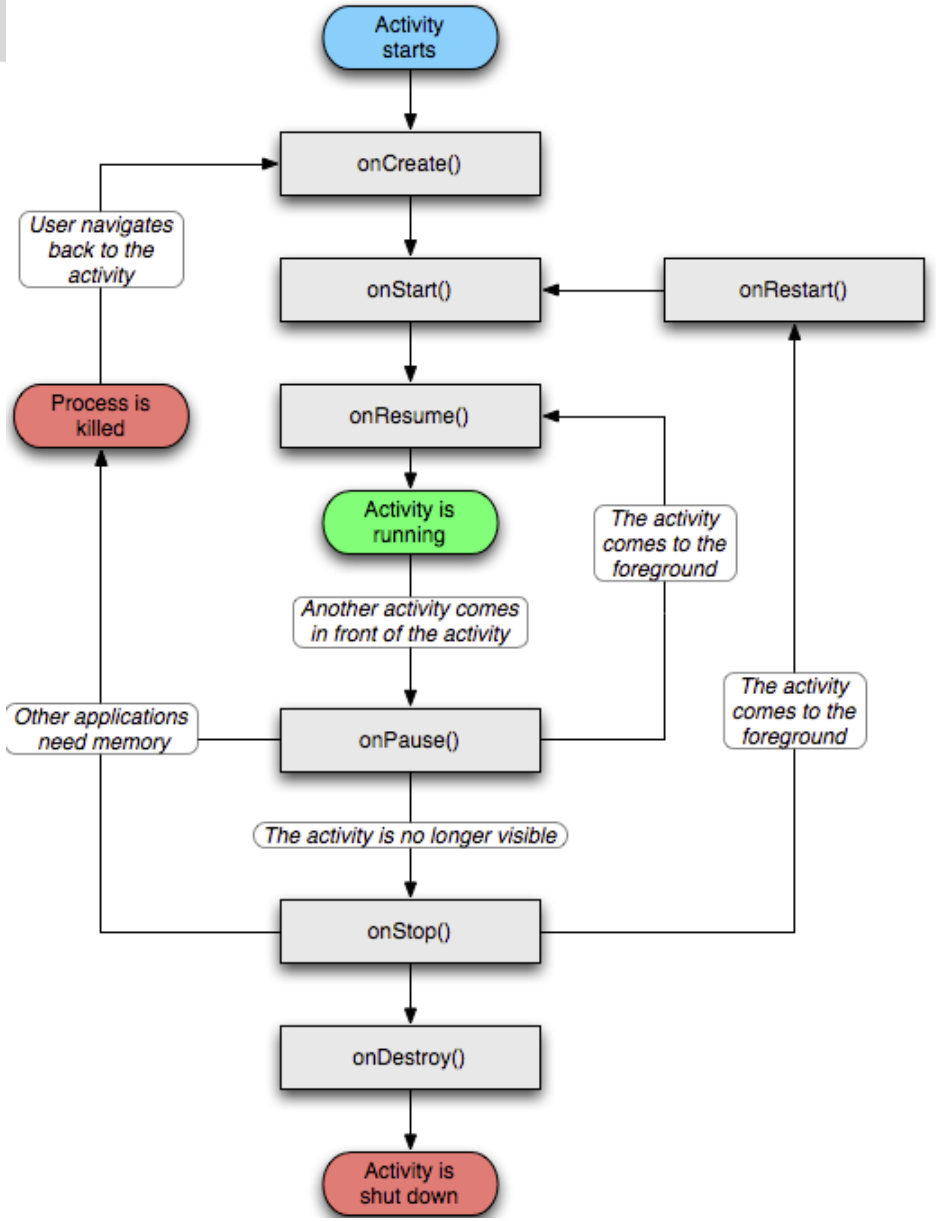
STARTING TASKS

- A task is created on executing an activity with filter action "**MAIN**" and category "**LAUNCHER**", this way the activity is accessible from the *desktop* apps manager. This is one way to retrieve the task once it has been sent to background, e.g. on pressing the HOME button.
- A long-press of the HOME button also provides access to recent apps.
- Otherwise, any task sent to background is unreachable.

ACTIVITIES AND TASKS

- The task contains a stack of activity objects, objects as instances of different or the same activity. The activities' stack is usually not reordered, so only **push** and **pop** operations are commonly used.
- A whole task can be brought back to the foreground (user application activated) or sent to background (user activity deactivated), for example, pressing the BACK and HOME buttons.

ACTIVITY CYCLE OF LIFE



RESOURCES

- They are external files used within the code and packed with the application.
- The resources must be located within the folder “res” and organized as follows:
  - res / drawable / png, jpg, 9.png and gif files
  - res / layout / screen layout XML files
  - res / values / predefined XML files, see next slide
  - Others…
- All file names hanging from res must be given in lower-case letters, otherwise you’ll have unexpected errors

RESOURCES

- res / values /
  - **arrays.xml** to declare arrays.
  - **colors.xml** to declare colors.
  - **dimens.xml** to declare dimensions.
  - **strings.xml** to declare strings of characters.
  - **styles.xml** to declare styles and themes.
  - Others…

DIMENSIONS

- Units to define dimensions:
  - dp, dip: density-independent pixels. These units are relative to a 160 dpi (dots per inch) screen
  - sp: scale-independent pixels: dip + scaled by the user's font size preference
  - pt: points (1/72 of an inch)
  - px: pixels
  - mm: millimeters
  - in: inches

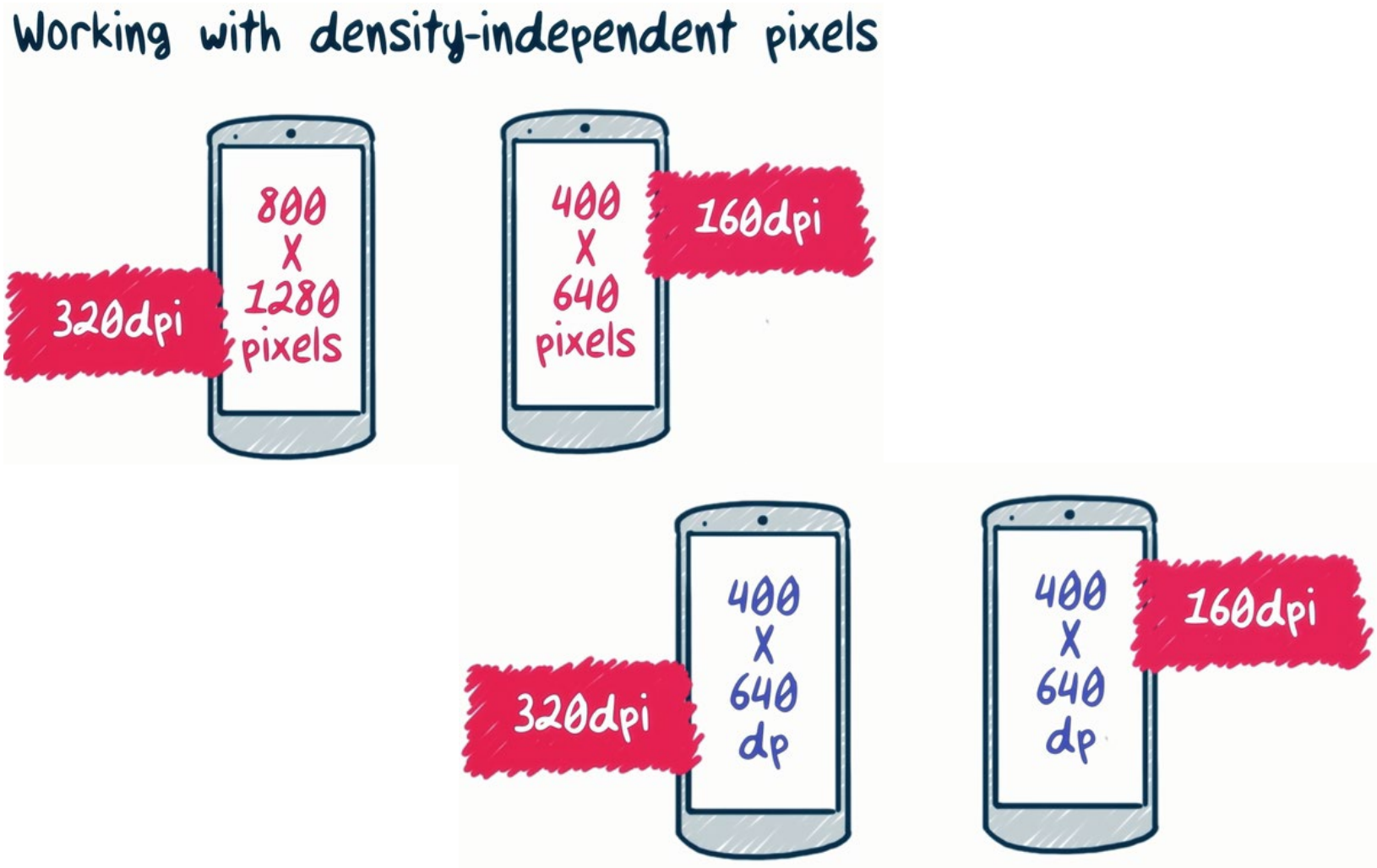
DIMENSION

Working with density-independent pixels

$$\frac{1 \text{ px}}{1 \text{ dp}} = \frac{160 \text{ dpi}}{160 \text{dpi}}$$

$$\frac{2 \text{ px}}{1 \text{ dp}} = \frac{320 \text{ dpi}}{160 \text{dpi}} \quad 2x$$

DIMENSIONES



RESOURCE QUALIFIERS

- The resources accessed by default are the ones defined on the previous slides
- It is possible to redefine the same resources (they must always be a subset of the resources given by default) for specific phone configurations or localizations, it is enough creating the same subfolders (named the same as the default folders) but adding appropriate **qualifiers**, always in-between hyphen marks.
- Look at the table on section “providing resources” from Android SDK for valid qualifiers. The naming using combinations of qualifiers must follow the precedence order defined also on the table.

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SUPPORTING DIFFERENT DEVICES

- Supporting Different Languages:

```
MyProject/
    res/
        values/
            strings.xml
        values-es/
            strings.xml
        values-fr/
            strings.xml
```

- Supporting Different Screens:

```
MyProject/
    res/
        layout/             # default (portrait)
            main.xml
        layout-land/        # landscape
            main.xml
        layout-large/       # large (portrait)
            main.xml
        layout-large-land/  # large landscape
            main.xml
```

- Create Different Bitmaps:

```
MyProject/
    res/
        drawable-xhdpi/
            awesomeimage.png
        drawable-hdpi/
            awesomeimage.png
        drawable-mdpi/
            awesomeimage.png
        drawable-ldpi/
            awesomeimage.png
```